

HOW TO FILL OUT THIS SHEET:



- Rub out all mistakes completely.
- Print your details clearly in the boxes provided.
- Make sure you fill in only one oval in each column.

EXAMPLE 1: Debbie Bach

FIRST NAME

D	E	B	B	I	E
A	A	A	A	A	A
B	B	B	B	B	B
C	C	C	C	C	C
D	D	D	D	D	D
E	E	E	E	E	E

LAST NAME

B	A	C	H
A	A	A	A
B	B	B	B
C	C	C	C
D	D	D	D
E	E	E	E

EXAMPLE 2: Chan Ai Beng

FIRST NAME **LAST NAME**

C	H	A	N
A	A		A
B	B		B
C	C		C
D	D		D
E			

A	I		B	E	N	G
	A	A	A	A		
	B	B	B	B		
	C	C	C	C		
	D	D	D	D		
E	E	E	E	E		

EXAMPLE 3: Jamal bin Abas

FIRST NAME										LAST NAME				
J	A	M	A	L			B	I	N	A	B	A	S	
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	
A	B	A	B	A	A	A	A	A	A	B	A	B	A	
B	B	B	B	B	B	B	B	B	B	B	B	B	B	
C	C	C	C	C	C	C	C	C	C	C	C	C	C	
D	D	D	D	D	D	D	D	D	D	D	D	D	D	
E	E	E	E	E	E	E	E	E	E	E	E	E	E	

FIRST NAME to appear on certificate

[illegible]

LAST NAME to appear on certificate

[illegible]

Are you male or female?

☐ Male ☐ Female



Does anyone in your home usually speak a language other than English?

☐ Yes ☐ No

School name: _____

Town / suburb: _____

Today's date: / / Postcode:

DATE OF BIRTH

Day		Month	Year	
0	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4		4	4	4
5		5	5	5
6		6	6	6
7		7	7	7
8		8	8	8
9		9	9	9

CLASS
(optional)

(A)	(K)
(B)	(L)
(C)	(M)
(D)	(N)
(E)	(O)
(F)	(P)
(G)	(Q)
(H)	(R)
(I)	(S)
(J)	(T)

TO ANSWER THE QUESTIONS

MULTIPLE CHOICE

Questions 1 to 40

Example: $4 + 6 =$

- (A) 2
- (B) 9
- (C) 10
- (D) 24

The answer is 10, so fill in the oval ☒, as shown.

☐ (A) ☐ (B) ☒ (C) ☐ (D)



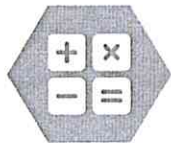
START

- 1 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 2 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 3 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 4 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 5 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 6 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 7 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 8 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 9 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 10 ☐ (A) ☐ (B) ☐ (C) ☐ (D)

- 11 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 12 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 13 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 14 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 15 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 16 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 17 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 18 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 19 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 20 ☐ (A) ☐ (B) ☐ (C) ☐ (D)

- 21 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 22 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 23 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 24 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 25 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 26 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 27 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 28 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 29 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 30 ☐ (A) ☐ (B) ☐ (C) ☐ (D)

- 31 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 32 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 33 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 34 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 35 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 36 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 37 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 38 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 39 ☐ (A) ☐ (B) ☐ (C) ☐ (D)
- 40 ☐ (A) ☐ (B) ☐ (C) ☐ (D)



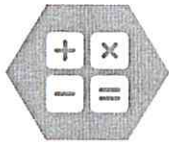
M

2012 Mathematics Answer Keys

ICAS
International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
1	B	A	D	D	D	C	A	B
2	B	A	C	A	B	A	C	B
3	A	D	B	C	A	C	C	D
4	A	D	B	B	B	A	B	B
5	D	B	A	A	D	B	B	B
6	D	D	D	C	D	B	A	A
7	D	C	A	B	A	B	A	A
8	A	B	B	D	B	A	D	C
9	B	D	A	C	C	A	C	D
10	D	B	A	D	D	D	B	B
11	B	B	C	A	B	A	C	A
12	C	B	B	A	A	B	B	C
13	C	A	A	D	B	D	D	D
14	C	A	B	B	A	D	C	C
15	D	C	D	D	C	C	B	B
16	A	A	C	D	B	C	C	B
17	A	C	C	B	D	B	C	C
18	A	C	D	A	C	A	A	D
19	B	B	D	C	A	D	A	A
20	A	B	A	B	D	C	C	B
21	D	D	C	B	B	A	D	A
22	C	A	A	D	C	C	C	A
23	B	C	D	B	C	C	B	D
24	D	B	B	B	B	B	B	C
25	C	C	B	A	A	D	B	D
26	C	A	D	C	A	C	D	A
27	A	C	C	A	C	C	B	D
28	B	C	A	B	B	A	B	C
29	B	D	C	A	B	C	D	A
30	A	B	C	C	B	C	A	C

(Please turn over)



M | 2012 Mathematics Answer Keys

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
31	C	C	C	A	C	B	D	D
32	C	D	D	A	C	D	B	D
33	A	C	A	C	C	B	D	D
34	A	B	A	B	A	D	A	C
35	B	C	D	D	A	D	D	A
36	C	D	D	72 072	5 05 005 0 5	16 016	4 04 004 0 4	26 026
37	D	A	B	10 010	600	566	3 03 003 0 3	9 09 009 0 9
38	D	A	C	90 090	32 032	27 027	4 04 004 0 4	39 039
39	C	C	C	5 05 005 0 5	9 09 009 0 9	137	9 09 009 0 9	75 075
40	D	B	B	500	900	6 06 006 0 6	127	126

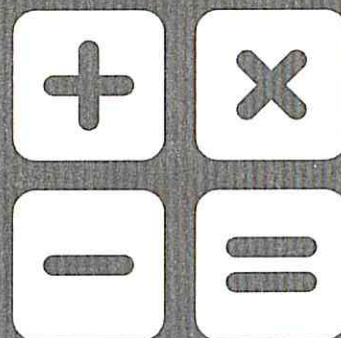
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Website: www.eaa.unsw.edu.au



UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

PAPER B



2012 ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

**Educational
Assessment
Australia**
eaa.unsw.edu.au

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).
Use the information provided to choose the **BEST** answer from the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.
You are **NOT** allowed to use a calculator.

1. Joe made this pattern.



What comes next in this pattern?



(A)



(B)

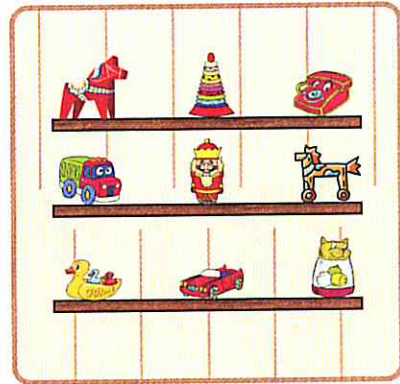


(C)



(D)

2. The teacher asked Lin to pick the toy on the right-hand side of the middle shelf.



Which toy must Lin pick?



(A)



(B)



(C)



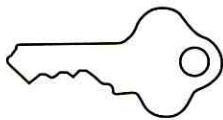
(D)

3. Mike had a bottle full of water. He poured all of the water into an empty glass. This filled the glass completely.

Which of these statements about the bottle and the glass **MUST** be true?

- (A) They have the same mass.
- (B) They have the same weight.
- (C) They have the same height.
- (D) They have the same capacity.

4. Sasi traced her key.



Which of these could be Sasi's key?

(A)



(B)



(C)



(D)



5. Fadi is planning a party for 16 people. He wants to buy party hats for them.

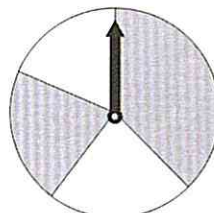


Party hats are sold in packets of five hats. Fadi wants everyone to have a party hat.

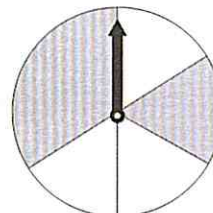
What is the smallest number of packets he must buy?

- (A) 3
(B) 4
(C) 11
(D) 16

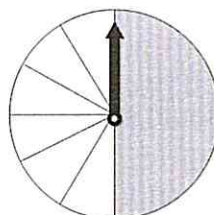
6. Which of the following spinners has more chance of landing on white than on grey?



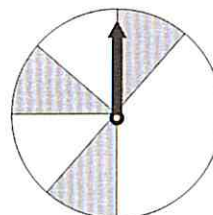
(A)



(B)

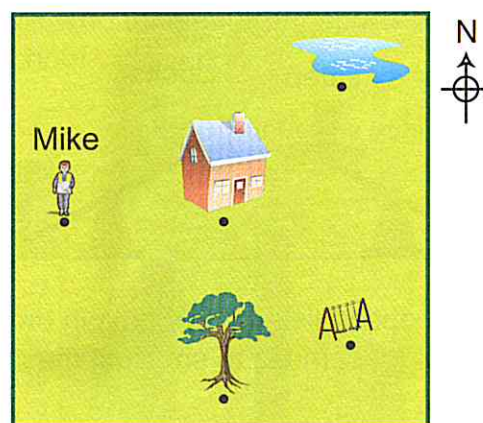


(C)



(D)

7. From where Mike is standing, he walks in a straight line to the tree.



In which direction does Mike walk?

- (A) north-east
(B) north-west
(C) south-east
(D) south-west

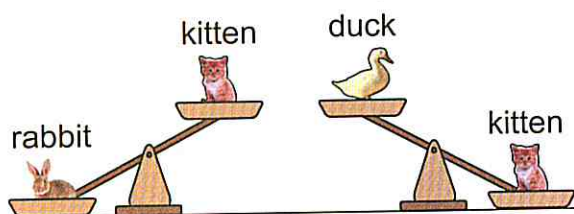
8. Ann has a ten-dollar note. She wants to buy an ice-cream that costs \$3.40.



How much change should she receive?

- (A) \$3.40
- (B) \$6.60
- (C) \$7.40
- (D) \$7.60

9. Mike is comparing the mass of his three pets.



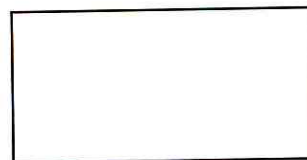
Which option lists Mike's pets in order from biggest mass to smallest mass?

- (A) duck, kitten, rabbit
- (B) kitten, rabbit, duck
- (C) rabbit, duck, kitten
- (D) rabbit, kitten, duck

10. This is a square centimetre.



How many square centimetres will cover this rectangle with no gaps or overlapping?



- (A) 6
- (B) 8
- (C) 10
- (D) 12

11. Class A, Class B and Class C all have some glass and plastic cups to use for science experiments.

The teacher counted the cups. She recorded the numbers in this table, but some values are missing.

Cup	Class A	Class B	Class C	Total
Glass	8	20		
Plastic		14	13	
Total	17		28	

How many cups did the teacher count?

- (A) 45
- (B) 79
- (C) 100
- (D) 158

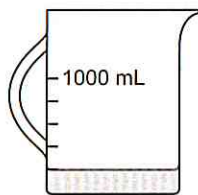
12. Lin has to solve this problem.

Find the number that is 75 less than 112, then double this number.

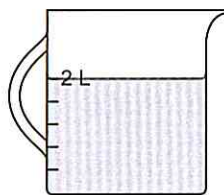
What is the answer to Lin's problem?

- (A) 37
- (B) 74
- (C) 86
- (D) 150

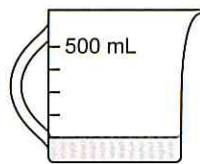
13. Which jug contains exactly 200 mL of liquid?



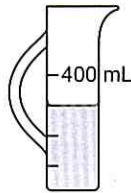
(A)



(B)



(C)



(D)

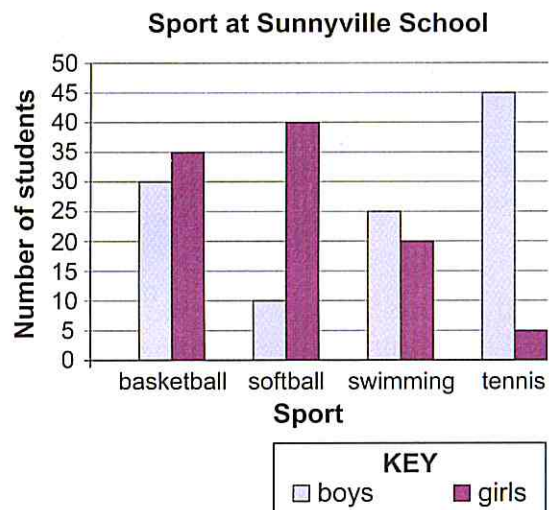
14. Lin exercises from 6:45 am to 7:10 am every day of the week.



For how long does Lin exercise each week?

- (A) 2 hours 55 minutes
- (B) 2 hours 15 minutes
- (C) 1 hour 45 minutes
- (D) 1 hour 25 minutes

15. This graph shows the number of boys and girls who take different sports at Sunnyville School.



Which sport is taken by the least number of students?

- (A) basketball
- (B) softball
- (C) swimming
- (D) tennis

16. Fadi drew this shape.



Which of these shows Fadi's shape flipped and turned?



(A)



(B)

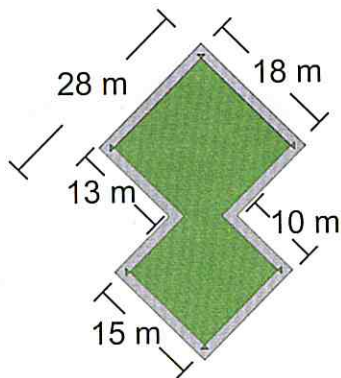


(C)



(D)

17. This diagram shows a fence around a garden. Some of the measurements are not shown.



What is the length of the fence?

- (A) 56 m
(B) 84 m
(C) 112 m
(D) 140 m

18. This is a number pattern.

209
228
247
266
285

?

323
342
361
380

What is the missing number?

- (A) 294
(B) 296
(C) 304
(D) 306

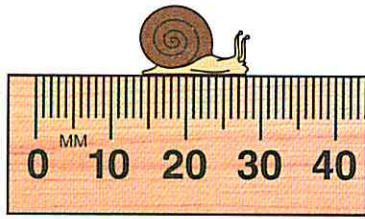
19. Joe wrote this calculation.

$$139 \times 6 - 134$$

Which of these will give the same answer?

- (A) $139 \times 7 - 5$
(B) $139 \times 5 + 5$
(C) $140 \times 5 + 5$
(D) $140 \times 6 - 128$

20. A snail was crawling along a ruler.



Fully stretched, the end of the snail's tail was on the 14 mm mark and its head just reached the 28 mm mark on the ruler.

Then the snail crawled to the right and stretched itself again. The end of its tail was now on the 18 mm mark.

What mark on the ruler did the snail's head reach?

- (A) 24 mm
- (B) 32 mm
- (C) 34 mm
- (D) 36 mm

21. Jake drew this picture of himself.



Which of these is a mirror image of Jake's picture?



(A)



(B)

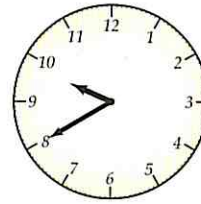


(C)

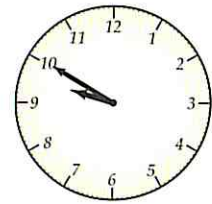


(D)

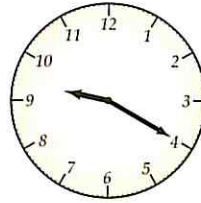
22. Which clock shows 10 minutes after 9:30?



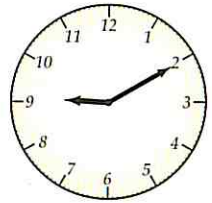
(A)



(B)



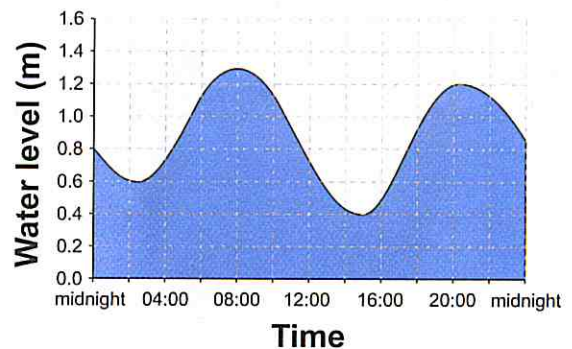
(C)



(D)

23. The chart shows how the water level in a harbour changed on a particular day.

Water Level on a Day



What was the difference between the highest and the lowest water level?

- (A) 0.1 m
- (B) 0.7 m
- (C) 0.9 m
- (D) 1.3 m

24. A class voted on whether they wanted a new school uniform.

This table shows the results.

	YES votes	NO votes
Boys	###	### ## I
Girls	### ##	###

What fraction of the girls did **not** want a new uniform?

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$
(C) $\frac{5}{11}$ (D) $\frac{5}{31}$

25. Luisa brought a basket of 30 small chocolate eggs to school.



She kept four eggs for herself.

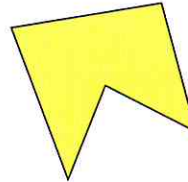
Then she gave two eggs to each of her friends.

She had six eggs left over.

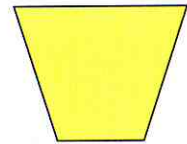
To how many friends did Luisa give the eggs?

- (A) 20
(B) 16
(C) 10
(D) 7

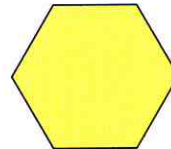
26. Which of these shapes is a pentagon?



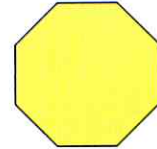
(A)



(B)

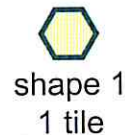


(C)

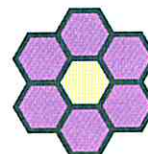


(D)

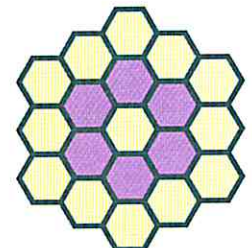
27. Arvind is making a pattern of shapes out of tiles.



shape 1
1 tile



shape 2
7 tiles



shape 3
19 tiles

How many tiles will be in shape 4?

- (A) 18
(B) 31
(C) 37
(D) 43

28. Sasi walks 600 steps from her house to reach her school.

Each of Sasi's steps measures 40 cm.

How far does Sasi walk to school?

- (A) 15 m
(B) 24 m
(C) 240 m
(D) 1500 m

29. Joe was given this subtraction.

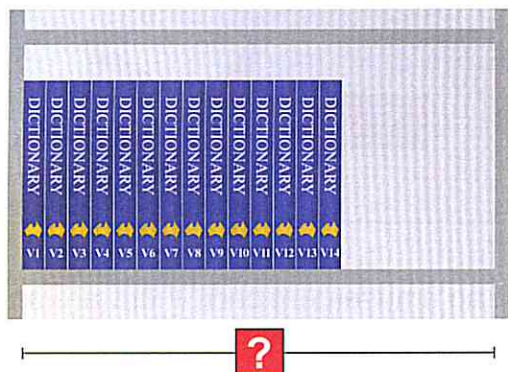
$$\begin{array}{r} 2 \quad ? \quad 6 \\ - 1 \quad 3 \quad 7 \\ \hline 1 \quad 1 \quad 9 \end{array}$$

What is the missing number?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

30. In the local library there is a set of 14 dictionaries on a shelf as shown.

The picture is drawn to scale.



Each dictionary has a width of 2.5 cm.

About how wide is the shelf?

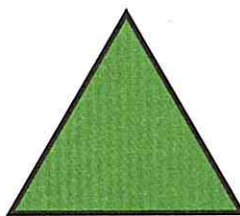
- (A) 35 cm
- (B) 50 cm
- (C) 60 cm
- (D) 70 cm

31. In Mrs Smith's class there are 10 boys and 11 girls. Each student brought 2 strawberries and 12 blueberries to a picnic.

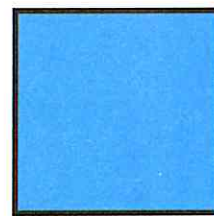
How many berries did the students bring altogether?

- (A) 35
- (B) 152
- (C) 294
- (D) 300

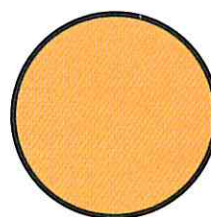
32. Which shape has NO lines of symmetry?



(A)



(B)



(C)



(D)

33. Sean had 24 coloured pencils.

He was given a new set of coloured pencils which contained twice as many as he already had.

Sean gave 11 coloured pencils to each of his three sisters.

How many pencils did he now have?

- (A) 36
- (B) 37
- (C) 39
- (D) 61

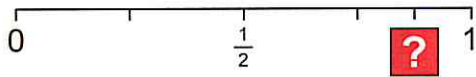
34. Mike thinks of two numbers.

When he adds the two numbers, the result is 15. When he multiplies the two numbers, the result is 54.

Which of these is the smaller of the two numbers?

- (A) 3
- (B) 6
- (C) 9
- (D) 12

35. What number should **?** be?



- (A) $\frac{4}{5}$ (B) $\frac{3}{4}$
(C) $\frac{7}{8}$ (D) $\frac{9}{10}$

36. Half a litre of milk fills 2 cups.

How many cups can be filled with 3 litres of milk?

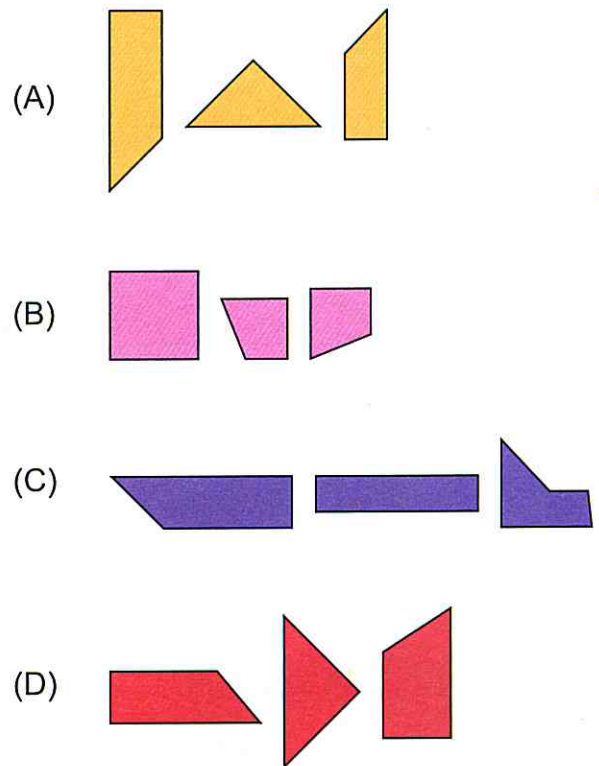
- (A) $1\frac{1}{2}$
(B) 3
(C) 6
(D) 12

37. Which card should be removed so the remaining cards make a pattern?



- (A) (B) (C) (D)

38. Which option shows three shapes that can make a rectangle when they are all joined?



39. At an amusement park, there are small rides and big rides.

Each small ride costs 3 tickets.
Each big ride costs 5 tickets.



Lin has 43 tickets.

What is the largest number of rides Lin can have if she uses **all** her tickets?

- (A) 9
(B) 11
(C) 13
(D) 14

40. This soccer ball is made of 12 black pentagonal panels and 20 white hexagonal panels.

There are 90 edges where the panels are joined together.



How many edges join two white panels?

- (A) 20
- (B) 30
- (C) 40
- (D) 60

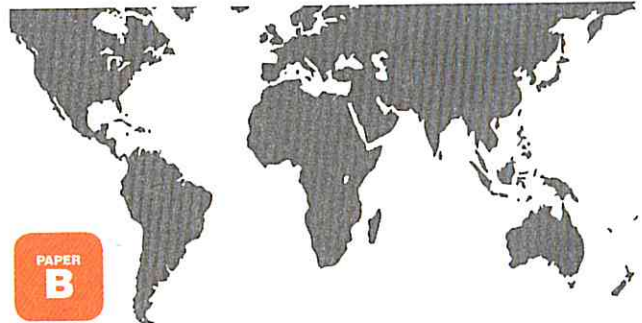
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The following year levels should sit THIS Paper:

Australia	Year 4
Brunei	Primary 4
Indonesia	Year 5
Malaysia	Standard 4
New Zealand	Year 5
Pacific	Year 4
Singapore	Primary 3
South Africa	Grade 4



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